ECE 2305 – Introduction to C Programming
Programming Project 01
Simple RC Circuit

A. Program Description:

The purpose of this program is to analyze the behavior of a simple RC circuit by computing the current and voltage across a capacitor over time. Given user input for time in microseconds, the program converts the input into seconds and calculates the corresponding values of current and voltage using exponential functions.

Inputs

- Time (t) in microseconds (entered by the user)

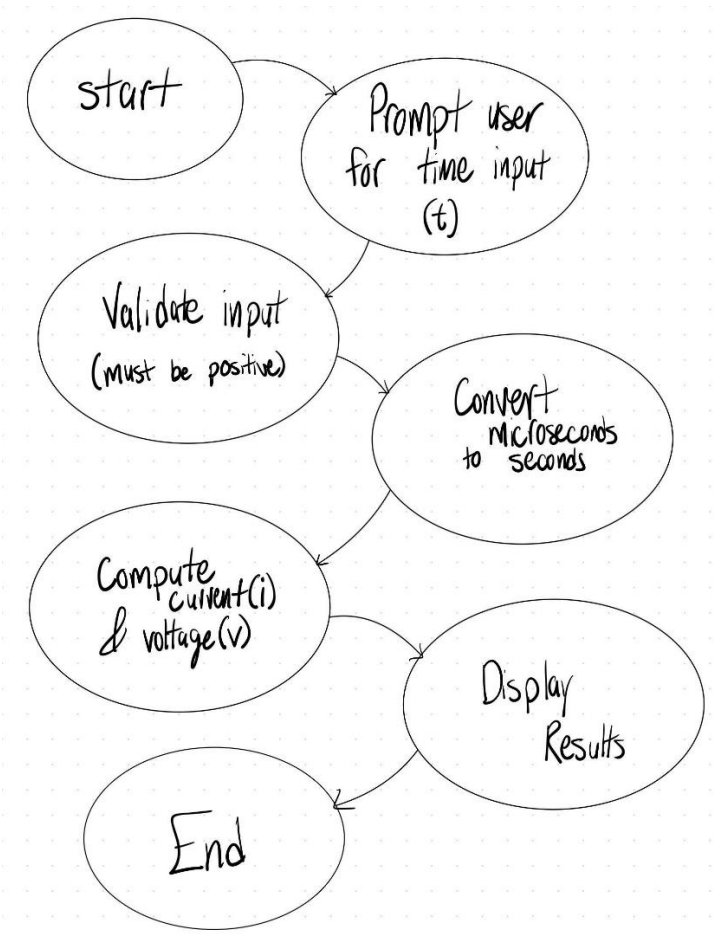
Outputs

- Current (i) in milliamps (mA)
- Voltage (v) in volts (V)

Program Structure

- The program uses sequential execution to define constants and read user input.
- It performs mathematical computations using predefined formulas.
- The final results are displayed using formatted output.

B. Program Flowchart:



C. Source Code

```cpp
#include <iostream>
#include <cmath>

using namespace std;

int main()
{
    // Constants
    const double V_DC = 10.0;
    const double R = 10000.0;
    const double C = 100.0e-12;

    double t_micro;  // Time in microseconds

    // Prompt user for input
    cout << "Enter time (t) in microseconds (us): ";
    cin >> t_micro;

    // Validate input (must be a number and positive)
    if (cin.fail() || t_micro < 0) {
        cout << "Invalid input. Please enter a positive numeric value." << endl;
        return 1;
    }

    // Convert microseconds to seconds
    double t = t_micro * 1e-6;

    // Compute current (i) in mA
    double i_t = (V_DC / R) * exp(-t / (R * C)) * 1000;

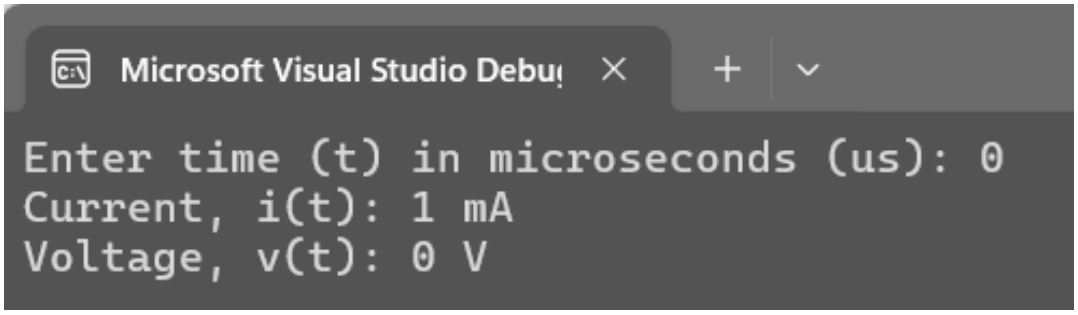
    // Compute voltage (v) in V
    double v_t = V_DC * (1 - exp(-t / (R * C)));

    // Display results
    cout << "Current, i(t): " << i_t << " mA" << endl;
    cout << "Voltage, v(t): " << v_t << " V" << endl;

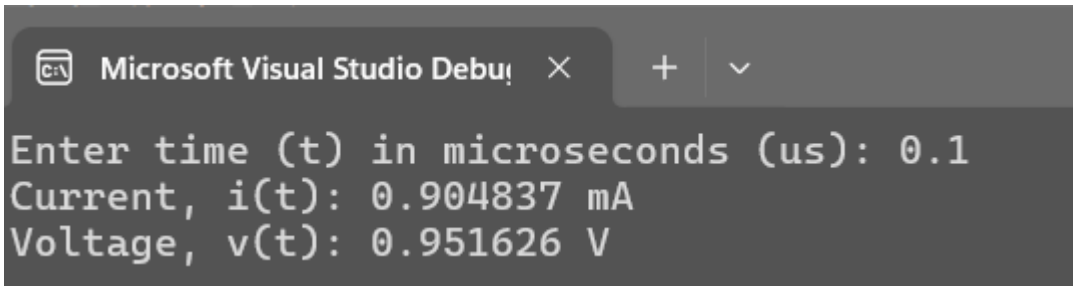
    return 0;
}
```

D. Program Test Results:

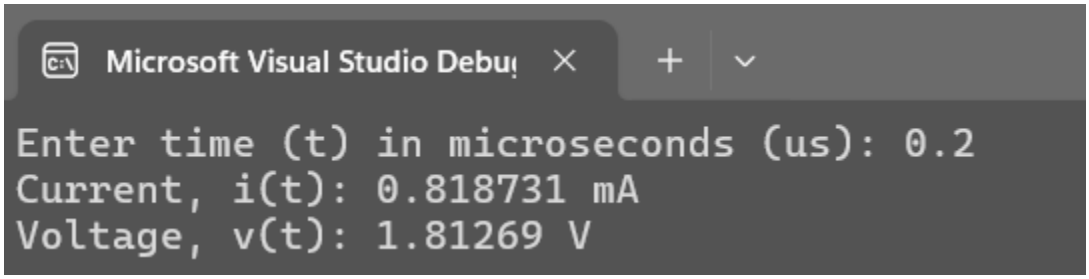
t = 0



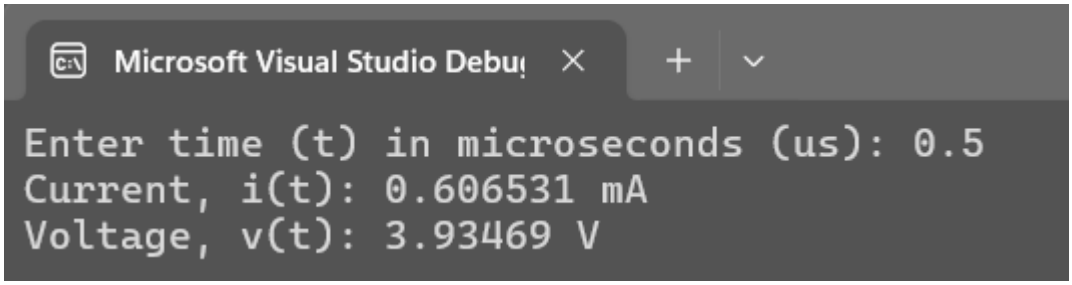
t = 0.1



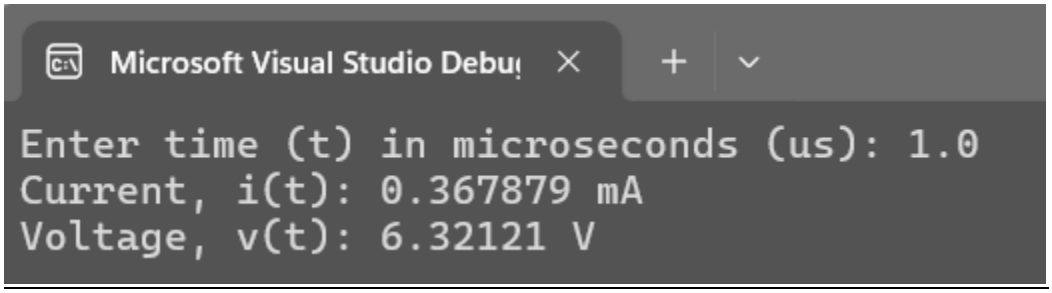
t = 0.2



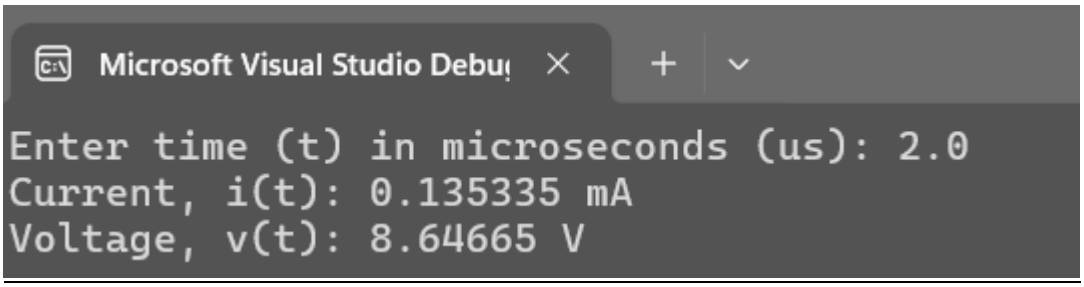
t = 0.5



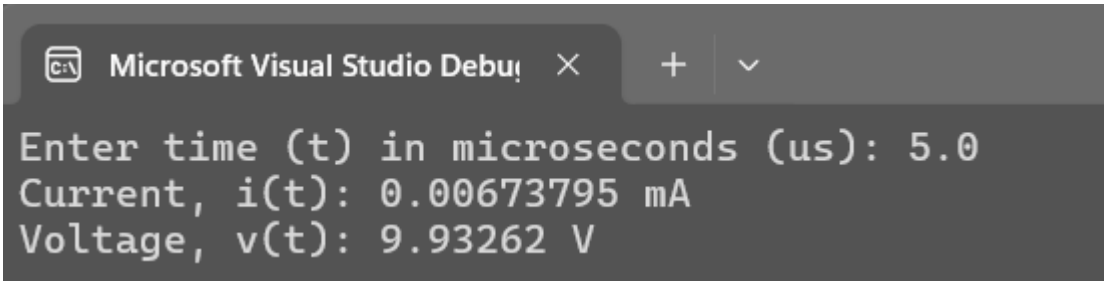
t = 1.0


Microsoft Visual Studio Debu
Enter time (t) in microseconds (us): 1.0
Current, i(t): 0.367879 mA
Voltage, v(t): 6.32121 V

t = 2.0


Microsoft Visual Studio Debu
Enter time (t) in microseconds (us): 2.0
Current, i(t): 0.135335 mA
Voltage, v(t): 8.64665 V

t = 5.0


Microsoft Visual Studio Debu
Enter time (t) in microseconds (us): 5.0
Current, i(t): 0.00673795 mA
Voltage, v(t): 9.93262 V

E. Tabulated Data.

Time (t) (μS)	Current, $i(t)$ (mA)	Voltage, $v(t)$ (V)
0	1 mA	0 V
0.1	0.904837 mA	0.951626 V
0.2	0.818731 mA	1.81269 V
0.5	0.606531 mA	3.93469 V
1.0	0.367879 mA	6.32121 V
2.0	0.135335 mA	8.64665 V
5.0	0.00673795 mA	9.93262 V